Elton Chang

(510) 513-2561 | trenan2004@gmail.com | https://eltonchang1.github.io

EDUCATION

Carnegie Mellon University May 2026
Master of Science in Data Analytics for Science, GPA: 3.97/4.0 Pittsburgh, PA

University of California, Santa Cruz Jun 2025
Bachelor of Science in Computer Science, GPA: 3.7/4.0 Santa Cruz, CA

EXPERIENCE

Founder / Technical Lead Aug 2025 - Present
Ashe System Pittsburgh, PA / San Francisco, CA

- Improved validated Precision@20 from 35% to 62% using XGBoost for temporal aircraft-on-ground (AOG) risk ranking across maintenance, inventory, and supplier features.
- Increased median warning lead time from 6.5 to 10.2 days using Optuna to optimize model hyperparameters and early-warning decision thresholds.

Infrastructure Engineer Intern Jul 2023 - Sep 2023
Wintec Industries Inc. Newark, CA

- Reduced configuration drift across more than 15 non-production hosts by using Terraform to provision standardized testing environments with reusable infrastructure modules.
- Cut anomaly detection time by 40% by engineering telemetry, alerting, and reliability dashboards across enterprise backup and replication pipelines.

Software Engineer Intern Jul 2022 - Sep 2022
Wintec Industries Inc. Newark, CA

- Built a cloud-hosted internal file management system that automated file ingestion with scheduled Python jobs, indexed metadata, and tracked company data lifecycle states.
- Improved onboarding and platform adoption by 35% by creating developer and user-facing documentation for architecture, workflows, and system usage.

PROJECTS

Next Best Action (Collaboration with PNC)

- Ranked high-impact advisor opportunities with a PNC Next Best Action pipeline using HistGradientBoosting, LightGBM, and a LightGBM-XGBoost ensemble to predict next-month product adoption.
- Achieved 0.811 ROC AUC and a 95% top-three hit rate on supported recommendations.

Torflix

- Built a BitTorrent-backed movie library with a FastAPI service for swarm I/O, storage, and progressive streaming, plus web clients for catalog, magnet enqueue, and playback while transfers complete.

TraceFrame

- Developed a local-first evidence tracker for data science workflows that records datasets, transformations, SQL outputs, metrics, charts, claims, checks, source-row samples, and lineage in a project-local .traceframe/ directory.

Quantifying Meritocracy (Research)

- Extended the University of Catania's Talent vs. Luck model with a solidarity parameter, using statistical modeling and agent-based simulations to quantify systemic drivers of inequality.

OPEN SOURCE CONTRIBUTIONS

mypy, Featuretools, SciPy, Narwhals, Panel, statsmodels, plotnine, Apache Airflow, SQLFluff, Shiny for Python, VegaFusion

TECHNICAL SKILLS

Python, TypeScript, C, Go, SQL, R, React, LangChain, LangGraph, LlamaIndex, Docker, Kubernetes